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#!/usr/bin/env python3 """ ShelbyFirst: Outcomes Data Table (standalone) Run: python shelby-
first_table_chart.py Output: src/data/processed/shelbyfirst_table_chart.png """
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```
import matplotlib matplotlib.use('Agg') import matplotlib.pyplot as plt from matplotlib.patches import Rect-
angle import numpy as np import pandas as pd import os from sklearn.linear_model import Ridge from
sklearn.preprocessing import StandardScaler from sklearn.pipeline import Pipeline
```

— Palette —

```
NAV = '#1b2a6b' TBL = '#2e4296' WHT = '#ffffff' EVN = '#eef1fa' GRY = '#dee3ef' RED = '#c0392b' GRN =
'#1a7a35' BLU = '#1a4db5' DRK = '#1a1a2e' SUB = '#8898cc'
```

— Model-driven projections —

```
_dir = os.path.dirname(os.path.abspath(file)) _data = os.path.join(_dir, 'data', 'processed')
_panel = pd.read_parquet(os.path.join(_data, 'enrichment_tract_year_panel.parquet')) _shelby =
_panel[_panel['geoid'].astype(str).str.startswith('47157')].copy() _shelby['income_k'] = _shelby['median_household_income']
/ 1000
_YEARS = [2024, 2025, 2026, 2027] UPTAKE = [0.0, 0.15, 0.35, 0.60]
_PEER = { 'labor_force': 66.9, 'poverty': 5.5, 'income': 119.0, 'insured': 94.0, 'no_veh': 4.2, 'dental': 32.1,
'obesity': 32.5, 'diabetes': 11.3, }
EFFECT = { 'labor_force': 0.22, 'poverty': 0.15, 'income': 0.10, 'insured': 0.40, 'no_veh': 0.20, 'dental': 0.35,
'obesity': 0.22, 'diabetes': 0.18, }
def _ridge_trend(df, col, scale=1.0): d = df[['year', col]].dropna() pipe = Pipeline([('sc', StandardScaler()),
('r', Ridge(alpha=1.0))]) pipe.fit(d[['year']].values, d[col].values) return (pipe.predict(np.array([[yr] for yr in
_YEARS])) * scale).tolist()
def _gap_boost(baseline, peer, pillar, direction): gap = abs(peer - baseline) eff = EFFECT[pillar] return
[baseline + direction * gap * eff * u for u in UPTAKE]
def _econ(df, col, scale, peer, pillar, direction): trend = _ridge_trend(df, col, scale) gap = abs(peer - trend[0])
eff = EFFECT[pillar] result = [trend[0]] for i in range(1, 4): result.append(trend[i] + direction * gap * eff *
UPTAKE[i]) return result
_lfpr_v = _econ(_shelby, 'lfpr_16p', 100, _PEER['labor_force'], 'labor_force', 1) _pov_v = _econ(_shelby,
'poverty_rate', 100, _PEER['poverty'], 'poverty', -1) _inc_v = _econ(_shelby, 'income_k', 1, _PEER['income'],
'income', 1)
_cs = pd.read_parquet(os.path.join(_data, 'cluster_shortlist_with_healthcare.parquet')) _r77 = _cs[_cs['rank']
== 11] _ins0 = (1 - float(_r77.iloc[0]['uninsured_rate'])) * 100
if len(_r77) and not pd.isna(_r77.iloc[0]['uninsured_rate']) else 82.0
_ins_v = _gap_boost(_ins0, _PEER['insured'], 'insured', 1) _noveh_v = _gap_boost(13.8, _PEER['no_veh'],
'no_veh', -1) _dental_v = _gap_boost(55.6, _PEER['dental'], 'dental', -1) _obes_v = _gap_boost(45.2,
_PEER['obesity'], 'obesity', -1) _diab_v = _gap_boost(21.6, _PEER['diabetes'], 'diabetes', -1)
_IGS = [35.8, 36.8, 38.4, 40.5]
METRICS = [('IGS Score', 1, _IGS, 55.7, '#4545c0', '', '{:.1f}'), ('Labor Force %', 1, _lfpr_v, _PEER['labor_force'],
'#1a8c5a', '', '{:.1f}%'), ('Insured %', 1, _ins_v, _PEER['insured'], '#c9a000', '', '{:.1f}%'), ('Med. Income',
1, _inc_v, _PEER['income'], '#e06000', '', '${:.1f}k'), ('Poverty %', -1, _pov_v, _PEER['poverty'], '#8b1515',
'', '{:.1f}%'), ('No-Veh HH %', -1, _noveh_v, _PEER['no_veh'], '#cc1515', '', '{:.1f}%'), ('Dental Gap %', -1,
_dental_v, _PEER['dental'], '#1a8cd4', '', '{:.1f}%'), ('Obesity %', -1, _obes_v, _PEER['obesity'], '#8bc820', '',
'{:.1f}%'), ('Diabetes %', -1, _diab_v, _PEER['diabetes'], '#30a040', '', '{:.1f}%'), ]
```

— Figure —

```
fig = plt.figure(figsize=(12, 9), dpi=130, facecolor=NAV)
```

— Header —

```
fig.text(0.030, 0.965, 'ShelbyFirst: Projected Outcomes', color=WHT, fontsize=22, fontweight='bold',
va='top') fig.text(0.030, 0.922, 'ShelbyFirst gap-closing model (Move · Nourish · Connect) · 'Uptake: 15%
Pilot → 35% Expand → 60% Scale · 96 tracts · pop. 273K', color=SUB, fontsize=10.5, va='top')
```

— Table —

```
ax_t = fig.add_axes([0.030, 0.115, 0.945, 0.775]) ax_t.set_xlim(0, 1) ax_t.set_ylim(0, 1) ax_t.axis('off')
COL_W = [0.280, 0.112, 0.108, 0.108, 0.108, 0.200] COL_X = list(np.cumsum([0.0] + COL_W[:-1])) COL_H
= ['Metric', 'Now (2024)', 'Year 1 2025', 'Year 2 2026', 'Year 3 2027', 'High IGS Shelby County']
ROW_H_HDR = 0.095 ROW_H_SEC = 0.070 ROW_H_DATA = 0.155
TABLE_ROWS = [ ('hdr', None), ('sec', '— IGS SCORE · Planning target — rises as a consequence of
ShelbyFirst programs —————'), ('data', METRICS[0]), ('sec', '— HEALTH METRICS ·
ShelbyFirst Move (rides → dental access) / Nourish (garden → obesity, diabetes)'), ('data', METRICS[6]),
('data', METRICS[7]), ('data', METRICS[8]), ]
y = 0.99 data_idx = 0
for rtype, rcontent in TABLE_ROWS: rh = {'hdr': ROW_H_HDR, 'sec': ROW_H_SEC, 'data': ROW_H_DATA}[rtype]
if rtype == 'hdr':
    for i, (cx, cw, ch) in enumerate(zip(COL_X, COL_W, COL_H)):
        ax_t.add_patch(Rectangle((cx, y - rh), cw, rh,
                                facecolor=NAV, edgecolor='#3a4ea0',
                                linewidth=0.6, zorder=2))
        xt = cx + 0.015 if i == 0 else cx + cw / 2
        fs = 10.5 if i == len(COL_H) - 1 else 13
        ax_t.text(xt, y - rh / 2, ch,
                  ha='left' if i == 0 else 'center', va='center',
                  fontsize=fs, color=WHT, fontweight='bold')
elif rtype == 'sec':
    ax_t.add_patch(Rectangle((0, y - rh), 1.0, rh,
                            facecolor=TBL, edgecolor='none', zorder=2))
    ax_t.text(0.015, y - rh / 2, rcontent,
              ha='left', va='center', fontsize=9,
              color='#aac8ff', style='italic')
else:
    tbl_lbl, direction, vals, peer, color, _, fmt = rcontent
    bg = EVN if data_idx % 2 == 1 else WHT
    data_idx += 1
    ax_t.add_patch(Rectangle((0, y - rh), 1.0, rh,
                            facecolor=bg, edgecolor=GRY,
                            linewidth=0.4, zorder=1))
```

```

# Left color accent bar
ax_t.add_patch(Rectangle((0, y - rh), 0.006, rh,
                        facecolor=color, edgecolor='none', zorder=3))

for cx in COL_X[1:]:
    ax_t.plot([cx, cx], [y - rh, y], color=GRY, linewidth=0.5, zorder=2)

ax_t.text(COL_X[0] + 0.018, y - rh / 2, tbl_lbl,
         ha='left', va='center', fontsize=14, color=DRK, fontweight='semibold')

ax_t.text(COL_X[1] + COL_W[1] / 2, y - rh / 2, fmt.format(vals[0]),
         ha='center', va='center', fontsize=14, color=RED, fontweight='bold')

ax_t.text(COL_X[2] + COL_W[2] / 2, y - rh / 2, fmt.format(vals[1]),
         ha='center', va='center', fontsize=14, color=DRK)

ax_t.text(COL_X[3] + COL_W[3] / 2, y - rh / 2, fmt.format(vals[2]),
         ha='center', va='center', fontsize=14, color=DRK)

ax_t.text(COL_X[4] + COL_W[4] / 2, y - rh / 2, fmt.format(vals[3]),
         ha='center', va='center', fontsize=15, color=GRN, fontweight='bold')

ax_t.text(COL_X[5] + COL_W[5] / 2, y - rh / 2, fmt.format(peer),
         ha='center', va='center', fontsize=14, color=BLU, fontweight='bold')

y -= rh

```

Footer

```

fig.text( 0.030, 0.012, 'Model: improvement = (peer gap) × program effectiveness × uptake · 'Econ met-
rics add Ridge year-trend as counterfactual baseline · 'Effectiveness grounded in Move (rides), Nourish
(garden), Connect (app/referrals) pillars · PDF slides 6–8', color=SUB, fontsize=8, va='bottom', )

```

Save

```

out = os.path.join(_dir, 'data', 'processed', 'shelbyfirst_table_chart.png') os.makedirs(os.path.dirname(out),
exist_ok=True) plt.savefig(out, dpi=130, bbox_inches='tight', facecolor=NAV, pad_inches=0.2) print(f'Saved:
{out}') plt.close()

```